T1 According to the title, this problem can be formulate as an integer problem:

$$\text{maximize} \quad 9x_1 + 4x_2 + 3x_3 + 2x_4 + \tfrac{1}{2}x_5$$

$$\text{s.t.} \quad 7x_1 + 5x_2 + 4x_3 + 3x_4 + x_5 \le 13.$$

$$x_1, x_2, x_3, x_4, x_5 \ge 0. \quad \text{integer}$$

And we can assume that $d[j]$ is the max value when the loaded weight is j. $(j = 0, 1, \cdots, 13)$

$d[0] = 0$, $d[j] = -\infty$, $j = 1, 2, \cdots, 13$.

$\therefore d[j] = \max\limits_{j \ge w_i} (d[j], d[j - w_i] + v_i)$

w_i is the weight of product i, v_i is the value of product i.

Let us consider product 1:

$w_1 = 7$, $v_1 = 9$

$d[7] = \max(d[7], d[0] + 9) = 9$

$d[13] = \max(d[13], d[6] + 9) = -\infty$.

Product 2:

$w_2 = 5$, $v_2 = 4$.

$d[5] = \max(d[5], d[0] + 4) = 4$

$d[6] = -\infty$, $d[7] = \max(d[7], d[2] + 4) = 9$,

$d[8] = -\infty$, $d[9] = -\infty$, $d[10] = \max(d[10], d[5] + 4) = 8$

$d[11] = -\infty$, $d[12] = \max(d[12], d[7] + 4) = 13$, $d[13] = -\infty$

Product 3.

$w_3 = 4$, $v_3 = 3$.

$d[4] = \max(d[4], d[0] + 3) = 3$

$d[5] = \max(d[5], d[1] + 3) = 4$

$d[6] = -\infty$, $d[7] = \max(d[7], d[3] + 3) = 9$

$d[8] = \max(d[8], d[4] + 3) = 6$, $d[9] = \max(d[9], d[5] + 3) = 7$

$d[10] = \max(d[10], d[6] + 3) = -\infty$, $d[11] = \max(d[11], d[7] + 3) = 12$

$d[12] = \max(d[12], d[8] + 3) = 13$, $d[13] = \max(d[13], d[9] + 3) = 10$

Product 4

$W_4=3, V_4=2$

$d[3]=\max(d[3], d[0]+2)=2$

$d[4]=\max(d[4], d[1]+2)=3$

$d[5]=\max(d[5], d[2]+2)=4$

$d[6]=\max(d[6], d[3]+2)=4$

$d[7]=\max(d[7], d[4]+2)=9$

$d[8]=\max(d[8], d[5]+2)=6$

$d[9]=\max(d[9], d[6]+2)=7$

$d[10]=\max(d[10], d[7]+2)=11$

$d[11]=\max(d[11], d[8]+2)=12$

$d[12]=\max(d[12], d[9]+2)=13$

$d[13]=\max(d[13], d[10]+2)=13.$

Product 5.

$W_5=1, V_5=0.5$

$d[1]=0.5$, $d[2]=\max(d[2], d[1]+0.5)=1$, $d[3]=\max(d[3], d[2]+0.5)=2$

$d[4]=3$, $d[5]=4$, $d[6]=4.5$, $d[7]=9$, $d[8]=9.5$, $d[9]=10$, $d[10]=11$, $d[11]=12$

$d[12]=13$, $d[13]=13.5$

As the result, choose one product 1, one product 2 and one product 5 is the optimal solution, the whole value is 13.5.

T2 According to the title, assume that $dp[i][j][k]$ represent the expected cost when the number of production from machine A is i, machine B is j, machine C is k.

$\therefore dp[0][0][0]=4\times 200=800.$

$$\therefore dp[i][j][k]=\min\begin{cases} dp[i-1][j][k]+Cost(A,1)-200B_1(i,j,k)+100C(i), \\ dp[i][j-1][k]+Cost(B,1)-200B_2(i,j,k)+300C(j), \\ dp[i][j][k-1]+Cost(C,1)-200B_3(i,j,k)+500C(k) \end{cases}$$

$B_1(i,j,k)=\max[(\max(0, 4-0.5(i-1)-0.8j-0.9k)-(4-0.5i-0.8j-0.9k)), 0]$

$B_2(i,j,k)=\max[(\max(0, 4-0.5i-0.8(j-1)-0.9k)-(4-0.5i-0.8j-0.9k)), 0]$

$B_3(i,j,k) = \max[(\max(0, 4-0.5i-0.8j-0.9(k-1)) - (4-0.5i-0.8j-0.9k)), 0]$

$Cost(A,i) = 20i$, $Cost(B,i) = 40i$, $Cost(C,i) = 60i$

$C(i) = \begin{cases} 1, & \text{if } i=1 \\ 0, & \text{if } i>1 \end{cases} = \max(0, 2-i)$